1. Explain the responsibilities of the data-link layer and distinguish logical link control from media access control.

The Data-Link Layer (Layer 2) is the second layer of the OSI model. It provides node-to-node delivery of data over a physical network link. It receives packets from the Network Layer and encapsulates them into frames.

Its main responsibilities are:

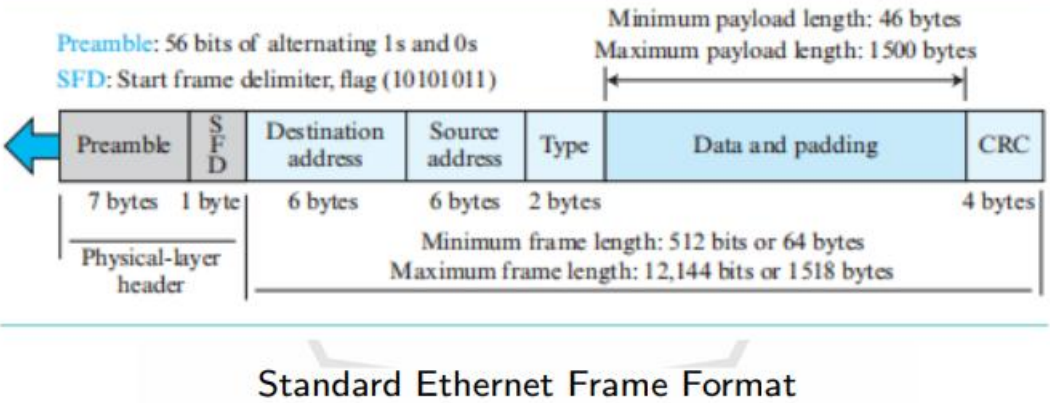
1. Framing: Divides network-layer packets into frames by adding headers and trailers.
2. Physical addressing: Uses MAC addresses to identify devices on a local network.
3. Error detection: Detects transmission errors using techniques such as Frame Check Sequence (FCS).
4. Flow control: Coordinates the rate of transmission where the particular link protocol supports it.
5. Media access control: Determines how devices share a common transmission medium.
6. Link management: Establishes, maintains, or terminates logical links where supported by the protocol.

The IEEE 802 data-link layer is commonly divided into two sublayers.

2. Explain Ethernet frame format and identify the purpose of destination MAC, source MAC, type/length and FCS fields.

Feature	**LLC (Logical Link Control)**	**MAC (Media Access Control)**
Position	Upper sublayer	Lower sublayer
Main function	Identifies and supports network-layer protocols	Controls access to the transmission medium
Addressing	Can identify the upper-layer protocol using SAP fields	Uses source and destination MAC addresses
Standard	IEEE 802.2	IEEE 802 family, such as 802.3 Ethernet and 802.11 Wi-Fi
Examples	Protocol multiplexing and logical link services	Ethernet frame addressing and Wi-Fi channel access

Figure 5.55 *Ethernet frame*



Standard Ethernet Frame Format

Preamble

- Size: 7 bytes = 56 bits.
- Contains alternating 0s and 1s.
- Alerts the receiver that a frame is coming.
- Helps the receiver synchronize its clock.
- Added at the Physical Layer.

Start Frame Delimiter (SFD)

- Size: 1 byte.
- Pattern: 10101011.
- Signals the beginning of the frame.
- Last two bits are 11.
- Also added at the Physical Layer.

Destination Address (DA)

- 6 bytes (48 bits).
- Identifies the destination station(s).

Source Address (SA)

- 6 bytes (48 bits).
- Identifies the sender.

Type

• Identifies the upper-layer protocol.

• Examples: IP, ARP, OSPF.

• Used for multiplexing/demultiplexing.

Data

• Carries the upper-layer packet.

• Minimum: 46 bytes.

• Maximum: 1500 bytes.

• Data below 46 bytes is padded.

CRC

• Last field of the frame.

• Uses CRC-32 for error detection.

• Corrupted frames are discarded.

3. Use Wireshark to capture Ethernet traffic and identify source and destination MAC addresses from at least five frames.

Aim

To capture Ethernet traffic and identify the source and destination MAC addresses from at least five frames.

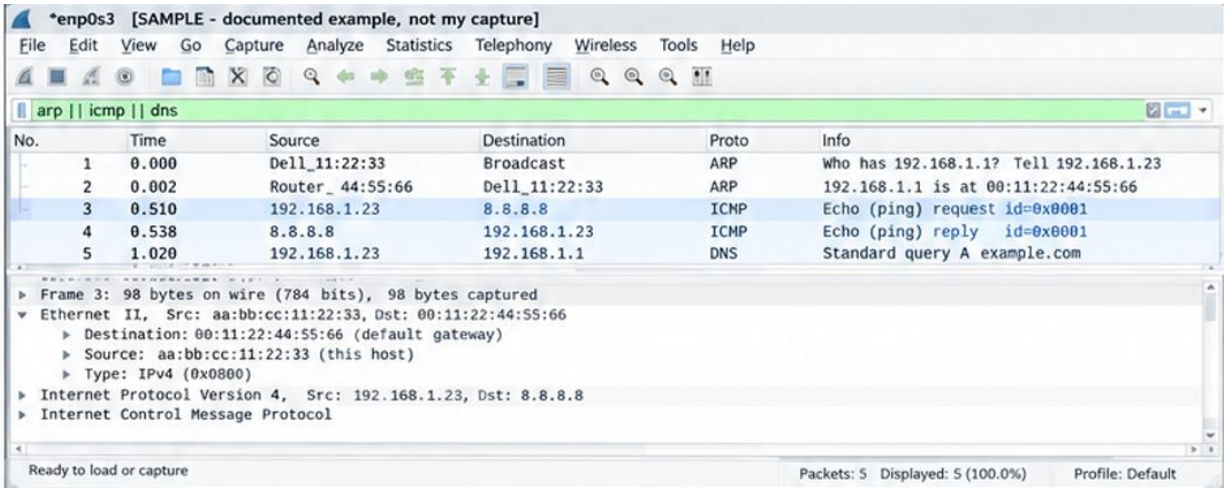
Requirements

- Wireshark installed on a computer.
- An active Ethernet connection, or a network interface that provides suitable traffic.
- Internet access or communication with another device.

Procedure

1. Open Wireshark on your computer.
2. Select the active Ethernet interface, such as Ethernet or eth0.
3. Double-click the interface to start capturing packets.
4. Generate traffic by opening a website or running ping to another device.
5. Stop the capture using the red stop button.
6. Enter the display filter eth to display Ethernet-related packets.
7. Select at least five frames and expand Ethernet II in the Packet Details pane.

8. Record the source MAC, destination MAC, and EtherType for each frame



Wireshark Example

Observation: The source MAC identifies the interface that transmitted the captured frame on the observed Ethernet link. The destination MAC identifies the intended local-link recipient or a broadcast/multicast destination.

Result: Wireshark can be used to inspect Ethernet frames and identify MAC addresses and protocol types.

4. Explain the operation of a switch using MAC address learning. Demonstrate how a switch builds its forwarding table.

A network switch is a Layer 2 device that forwards Ethernet frames using a MAC address table, also known as a CAM table or forwarding table.

The switch follows four main rules:

1. Learning: It reads the source MAC address of an incoming frame and associates that address with the port on which the frame arrived.
2. Forwarding: If the destination MAC address is known, it forwards the frame only through the corresponding port.
3. Flooding: If the destination address is unknown, or the frame is a broadcast, it forwards the frame to other eligible ports in the same VLAN, except the incoming port.
4. Filtering: If the destination is already associated with the incoming port, the switch does not forward the frame to other ports.

Consider four computers connected to a new switch:

- Computer A → Port 1

- Computer B → Port 2
- Computer C → Port 3
- Computer D → Port 4

Initially, the MAC address table is empty.

Step	Frame transmitted	Switch operation	MAC table after the frame
1	A → B	Learns A on port 1; B is unknown, so floods to ports 2, 3 and 4	A:1
2	B → A	Learns B on port 2; forwards to port 1	A:1, B:2
3	C → A	Learns C on port 3; forwards to port 1	A:1, B:2, C:3
4	D → B	Learns D on port 4; forwards to port 2	A:1, B:2, C:3, D:4
5	A → C	Destination is known; forwards only to port 3	Unchanged

5. Compare hub, switch and bridge in terms of forwarding, collision domains and practical use.

Feature	Hub	Switch	Bridge
OSI layer	Layer 1	Layer 2	Layer 2
Forwarding method	Repeats incoming signals to other ports	Forwards frames using MAC addresses	Forwards frames using MAC addresses
MAC learning	No	Yes	Yes, in learning bridges
Collision domains	One shared collision domain	One per port in traditional Ethernet switching	Separates collision domains between bridged segments
Broadcast domain	One	Usually one per VLAN	Usually one across bridged segments
Duplex	Typically half-duplex	Full-duplex commonly supported	Depends on implementation
Performance	Low	High	Generally lower port density than modern switches

Practical use	Legacy networks and demonstrations	Modern LANs	Connecting or segmenting LAN segments

6. Explain CSMA/CD and discuss why its importance is reduced in modern switched full-duplex Ethernet.

CSMA/CD stands for Carrier Sense Multiple Access with Collision Detection. It was used in traditional shared Ethernet networks.

Working procedure

1. Carrier sense: A station listens to the medium to check whether another station is transmitting.
2. Multiple access: Several stations share the same communication medium.
3. Transmit: If the medium is idle, the station starts transmitting.
4. Collision detection: During transmission, the station monitors for a collision.
5. Jam signal: If a collision is detected, the station stops transmitting and sends a jam signal according to the protocol.
6. Backoff: The station waits for a randomly selected interval using binary exponential backoff before trying again.

Example

Two computers connected to a shared, half-duplex Ethernet segment may sense the medium as idle at nearly the same time. Both transmit, causing a collision. They stop, wait for different backoff intervals, and retry.

Why is CSMA/CD less important today?

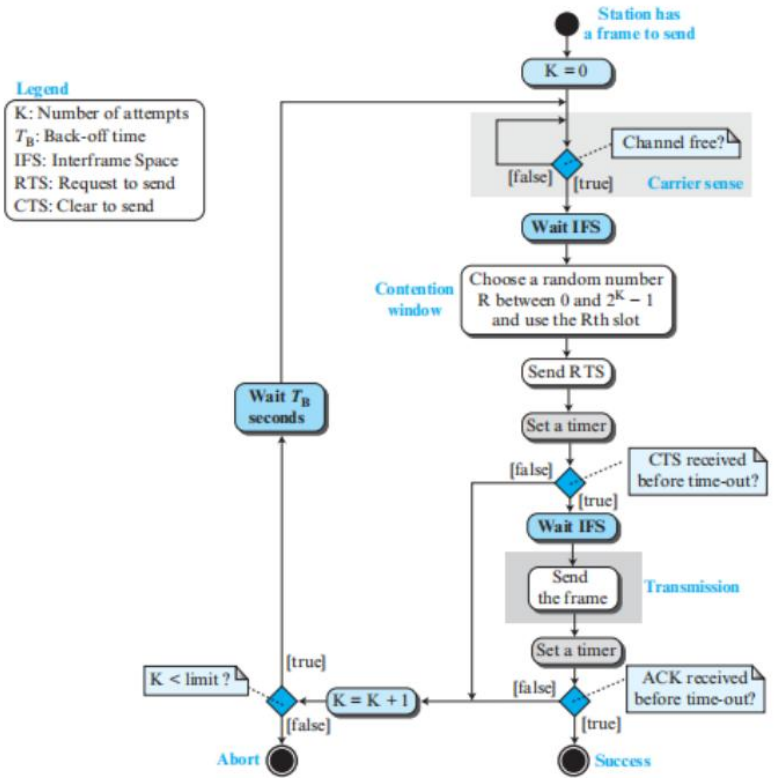
- Modern Ethernet commonly uses switches rather than shared hubs.
- Each switched port normally has its own link to a device.
- Full-duplex Ethernet permits simultaneous transmission and reception.
- Full-duplex links do not use collision detection because collisions are not expected on these links.
- Therefore, CSMA/CD is mainly relevant to legacy shared and half-duplex Ethernet.

7. Explain CSMA/CA in IEEE 802.11 and compare its objective with CSMA/CD.

CSMA/CA stands for Carrier Sense Multiple Access with Collision Avoidance. It is used by Wi-Fi networks defined by the IEEE 802.11 family of standards.

Unlike traditional shared Ethernet, wireless devices cannot always detect a collision while transmitting. CSMA/CA therefore tries to reduce the probability of collisions before transmission begins.

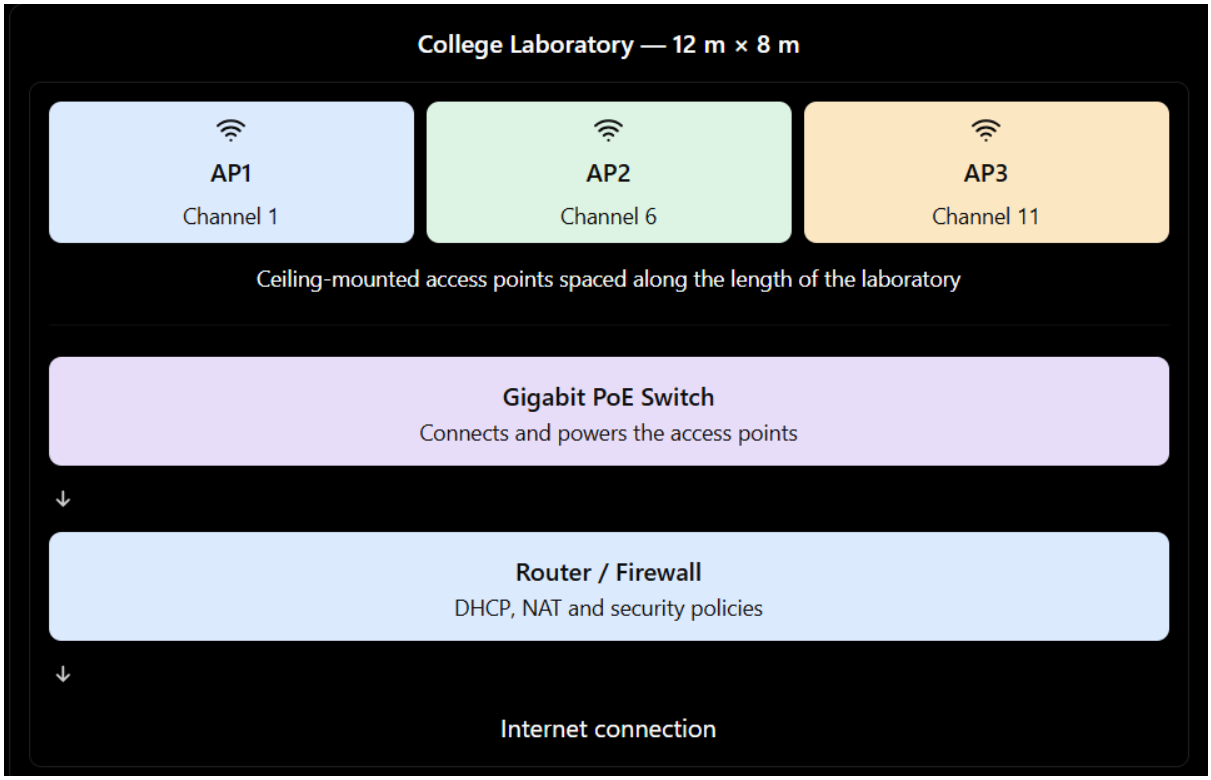
Figure 6.7 *Flow diagram of CSMA/CA*



Feature	CSMA/CD	CSMA/CA
Full form	Collision Detection	Collision Avoidance
Common use	Traditional shared half-duplex Ethernet	Wi-Fi / IEEE 802.11
Main objective	Detect collisions and recover	Reduce the likelihood of collisions
Collision handling	Detects collisions during transmission	Uses channel sensing, random backoff and acknowledgments
Acknowledgments	Not used by classic Ethernet MAC for each frame	MAC ACKs are commonly used for unicast frames
Modern relevance	Mostly legacy Ethernet	Important in shared wireless communication

8. Design a small wireless LAN for a college laboratory. Specify access point placement, SSID, channel considerations, addressing and basic security requirements.

To design a wireless LAN for a college laboratory measuring 12 m × 8 m, containing approximately 40 workstations and additional laptops, with around 60 devices connected simultaneously.



Design specifications

Component	**Proposed configuration**
Laboratory size	12 m × 8 m
Number of access points	3
Placement	Ceiling-mounted at approximately 2.7 m, spaced along the room
Cell overlap	Approximately 15–20%, adjusted after a coverage survey
Wi-Fi standard	Wi-Fi 5 or Wi-Fi 6, dual-band
Student/staff SSID	College-Lab
Guest SSID	College-Guest

2.4 GHz channels	Channels 1, 6 and 11 on AP1, AP2 and AP3, respectively, where permitted and suitable
5 GHz channels	Examples include 36, 40 and 44, subject to local regulations and channel planning
Backbone	Gigabit PoE switch connected to a router/firewall
Capacity target	Approximately 20 clients per AP as an initial planning target

9. Explain Mobile IP and the need for mobility support when a device moves between networks.

Mobile IP is a networking approach that allows a mobile device to retain a stable home IP address while moving between networks, with support from the relevant mobility protocol.

Ordinary IP routing generally depends on the device's current network prefix. If a device moves to another IP network, it may receive a new IP address, potentially interrupting existing communications.

Mobile IP was developed to address this problem.

- Mobile node: A device that moves between networks.
- Home network: The network associated with the device's permanent home address.
- Home address: The stable IP address associated with the mobile node.
- Foreign network: A network currently visited by the device.
- Care-of address (CoA): An address representing the mobile node's current point of attachment.
- Home agent (HA): An agent on the home network that helps locate and forward traffic to the mobile node.
- Correspondent node: The device communicating with the mobile node.

Example

Suppose a mobile device has a home address of 192.168.1.20. It moves to another network and obtains a care-of address of 10.0.0.25.

With Mobile IPv4, a correspondent can continue addressing packets to the home address. The home agent can intercept those packets and tunnel them toward the care-of address.

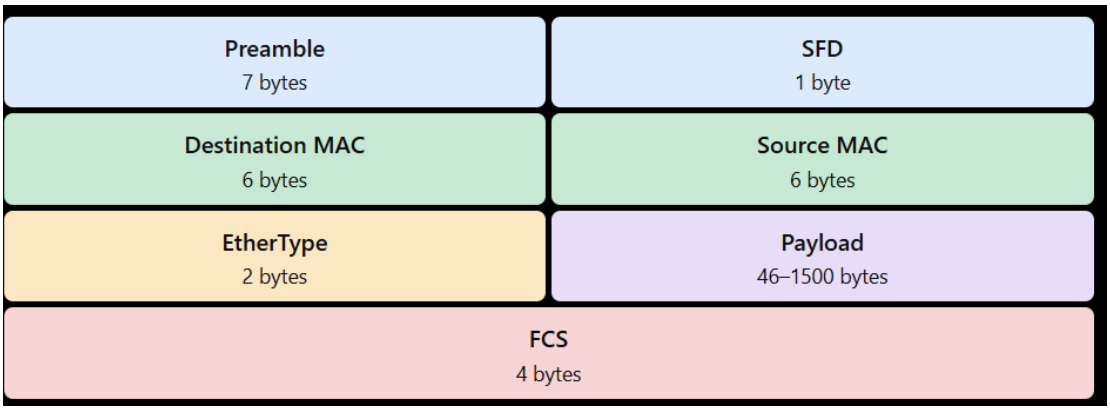
This allows the device to remain reachable using its home address, subject to correct configuration and network support.

Why is mobility support necessary?

1. Session continuity: Helps maintain communication when the device changes networks.
2. Stable addressing: Allows the device to retain its home address.
3. Reachability: Enables correspondents to locate a mobile device after it moves.
4. Transparent movement: Reduces the need for correspondents to know the device's current location.

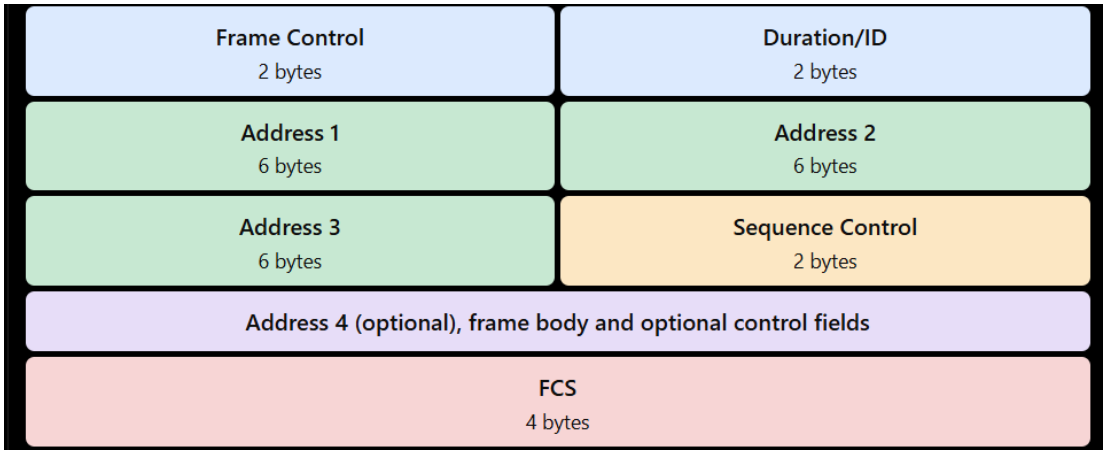
10. Perform a packet-level comparison of a wired Ethernet frame and a wireless LAN frame using suitable captures or documented examples.

Wired Ethernet frame



Wireless LAN frame

A typical three-address IEEE 802.11 data frame has a 24-byte MAC header. A four-address frame commonly has a 30-byte header before any additional optional fields.



Field	Wired Ethernet II	Wireless LAN (802.11)

Frame identification	Destination/source MAC, Type/Length	Frame Control
Addressing	Two MAC addresses in the basic header	Typically three or four MAC address fields, depending on frame type
Header size	14 bytes without VLAN tag	Common three-address MAC header: 24 bytes; four-address frames commonly use 30 bytes
Payload identification	EtherType identifies IPv4, IPv6, etc.	Frame type/subtype indicates management, control or data; LLC/SNAP may identify the payload protocol
Error detection	4-byte FCS	4-byte FCS
Medium access	Full-duplex switched Ethernet normally has no collisions	Channel sensing, random backoff and acknowledgment mechanisms
Additional fields	Optional VLAN tag and other extensions	May include QoS Control, HT Control and other fields